Student Results Management System

Task 2 | Console-Based JavaScript | Classes & Objects + map, filter, reduce

Requirements Checklist

#	Requirement	Implementation
1	Class & Constructor	Student(id, name, grades, attendance)
2	Analysis methods	reduce for average, if/else for grade, && for result
3	Objects array	studentsList with 4 students (2 struggling)
4	Leniency committee	filter -> map (+5) -> reduce, with zero-division shield
5	Console logging	Student reports, leniency array, new average

Source Code

```
// ============================================================
// Task 2: Student Results Management & Analysis System (Console)
// Concepts: Classes/Objects + map, filter, reduce
// ============================================================

// ---------- 1. Class & Constructor ----------
class Student {
  constructor(id, name, grades, attendance) {
    this.id = id;                   // seat number
    this.name = name;               // student name
    this.grades = grades;           // array of 3 subject grades
    this.attendance = attendance;   // attendance percentage
  }

  // ---------- 2. Academic analysis methods ----------

  // Average using reduce
  calculateAverage() {
    const total = this.grades.reduce((sum, grade) => sum + grade, 0);
    return total / this.grades.length;
  }

  // Letter grade using if/else
  getGrade() {
    const avg = this.calculateAverage();
    if (avg >= 90) {
      return "A";
    } else if (avg >= 80) {
      return "B";
    } else if (avg >= 70) {
      return "C";
    } else if (avg >= 60) {
      return "D";
    } else if (avg >= 50) {
      return "E";
    } else {
      return "F";
    }
  }

  // Final result using && (both conditions must be true)
```

```
  getFinalResult() {
    const avg = this.calculateAverage();
    if (avg >= 50 && this.attendance >= 75) {
      return "ناجح";
    } else {
      return "يحتاج مراجعة";
    }
  }
}

// ---------- 3. Main objects array ----------
const studentsList = [
  new Student(1001, "Ahmed Ali",    [85, 90, 78], 92),
  new Student(1002, "Sara Hassan",  [95, 88, 92], 96),
  new Student(1003, "Omar Khaled",  [45, 42, 48], 80),  // struggling (avg 45)
  new Student(1004, "Mona Ibrahim", [40, 47, 44], 70)   // struggling (avg 43.67)
];

// ---------- 4. Leniency committee (Array Methods) ----------

// Extraction: filter students whose average is between 40 and 49
const strugglingStudents = studentsList.filter((student) => {
  const avg = student.calculateAverage();
  return avg >= 40 && avg <= 49;
});

// Processing: map to add 5 leniency points to their average
const leniencyStudents = strugglingStudents.map((student) => {
  return {
    id: student.id,
    name: student.name,
    oldAverage: student.calculateAverage(),
    newAverage: student.calculateAverage() + 5
  };
});

// Analysis: reduce to sum the new averages
const totalNewAverages = leniencyStudents.reduce(
  (sum, student) => sum + student.newAverage,
  0
);

// Leniency average with a safety shield (avoid division by zero)
const leniencyAverage =
  leniencyStudents.length === 0
    ? 0
    : totalNewAverages / leniencyStudents.length;

// ---------- 5. Console logging ----------

console.log("========== Student Reports ==========");
studentsList.forEach((student) => {
  console.log(
    `Name: ${student.name} | Average: ${student.calculateAverage().toFixed(2)} | ` +
    `Grade: ${student.getGrade()} | Result: ${student.getFinalResult()}`
  );
});

console.log("\n========== Students Under Leniency ==========");
console.log(leniencyStudents);

console.log("\n========== Leniency Average (after adjustment) ==========");
console.log(leniencyAverage.toFixed(2));
```

Console Output

```
========== Student Reports ==========
Name: Ahmed Ali | Average: 84.33 | Grade: B | Result: ناجح
Name: Sara Hassan | Average: 91.67 | Grade: A | Result: ناجح
Name: Omar Khaled | Average: 45.00 | Grade: F | Result: يحتاج مراجعة
Name: Mona Ibrahim | Average: 43.67 | Grade: F | Result: يحتاج مراجعة

========== Students Under Leniency ==========
[
  { id: 1003, name: 'Omar Khaled', oldAverage: 45, newAverage: 50 },
  {
    id: 1004,
    name: 'Mona Ibrahim',
    oldAverage: 43.666666666666664,
    newAverage: 48.666666666666664
  }
]

========== Leniency Average (after adjustment) ==========
49.33
```